

AI Agent Evaluation Metrics — Reference Table

Based on the supplied AI Agent Evaluation guide and its DeepEval material. Thresholds shown as examples are document examples, not universal recommendations.

Metric	What it checks	Calculation	Threshold shown
Plan Quality	Is the plan good and aligned with the task?	$\text{AlignmentScore}(\text{Task}, \text{Plan})$	Not explicitly specified
Plan Adherence	Did the agent follow its own plan?	$\text{AlignmentScore}((\text{Task}, \text{Plan}), \text{Execution Steps})$	Not explicitly specified
Tool Correctness	Did it choose the right tool?	Correctly Used Tools / Total Tools Called	0.5 default
Argument Correctness	Are the tool inputs correct?	Correctly Generated Input Parameters / Total Tool Calls	0.5 default; 0.7 in example
Task Completion	Did the agent finish the user's task?	$\text{AlignmentScore}(\text{Task}, \text{Outcome})$	0.7 in example
Step Efficiency	Did it take an efficient path?	$\text{AlignmentScore}(\text{Task}, \text{Execution Steps})$	0.7 in example

General threshold rule in the source: $\text{Score} \geq \text{Threshold} \rightarrow \text{Pass}$; $\text{Score} < \text{Threshold} \rightarrow \text{Fail}$. The guide also notes that strict mode changes the interpretation to 1 = perfect and 0 = otherwise.

Metric Explanations and Calculations

The following explanations simplify the descriptions in the supplied PDF while keeping the formulas and evaluation logic given in the source.

1. Plan Quality (PlanQualityMetric)

This checks whether the agent's plan is a good plan for the user's task. In simple words: **Did the agent choose a sensible strategy before doing the work?**

Formula / calculation:

$$\text{Plan Quality Score} = \text{AlignmentScore}(\text{Task}, \text{Plan})$$

How is it calculated?

1. Take the user's task from the trace.
2. Extract the plan from the agent's reasoning.
3. Check whether the plan is relevant, logical, complete, and aligned with the task.
4. An LLM generates the alignment score.

Simple example: Task: Find AI/ML Engineer jobs in Hyderabad. A good plan searches relevant jobs, filters for Hyderabad, ranks them, and returns recommendations. A poor plan includes unrelated work.

How to interpret the score: Higher score means a better plan. A low score means the strategy itself is poor or contains unnecessary/unrelated actions.

2. Plan Adherence (PlanAdherenceMetric)

This checks whether the agent actually follows the plan it created. In simple words: **The plan was good, but did the agent stick to it?**

Formula / calculation:

$$\text{Plan Adherence Score} = \text{AlignmentScore}((\text{Task}, \text{Plan}), \text{Execution Steps})$$

How is it calculated?

1. Extract the user's task.
2. Extract the agent's plan.
3. Look at the actual execution steps.
4. Compare the execution with the task and planned strategy.
5. An LLM generates the final alignment score.

Simple example: Plan: Search → Filter Hyderabad → Rank → Return. If the agent searches unrelated information and generates a cover letter before ranking, it has deviated from the plan.

How to interpret the score: Higher score means the agent stayed closer to its planned strategy. Low adherence means execution deviated.

3. Tool Correctness (ToolCorrectnessMetric)

This checks whether the agent selected the correct tool. In simple words: **Did it use the right tool?**

Formula / calculation:

$$\text{Tool Correctness} = \text{Correctly Used Tools} / \text{Total Tools Called}$$

How is it calculated?

1. Record the tools the agent actually called.
2. Compare them with the expected tools.

3. The evaluation can consider tool names, input parameters, outputs, ordering, and exact matching when configured.
4. If `available_tools` is supplied, an LLM can judge whether the selected tools were optimal.
5. The source states that the final score can be the minimum of the deterministic score and the LLM-based optimality score.

Simple example: If the task is to find jobs and the agent calls a job-search tool, that is appropriate. Calling a weather-search tool would indicate incorrect tool selection.

How to interpret the score: The PDF gives a default threshold of 0.5. $\text{Score} \geq 0.5$ passes under that default. Strict mode uses 1 = perfect and 0 = otherwise.

4. Argument Correctness (ArgumentCorrectnessMetric)

This checks whether the agent gave the correct information to the tool. In simple words: **The tool is right, but did the agent give it the right inputs?**

Formula / calculation:

Argument Correctness = Correctly Generated Input Parameters / Total Number of Tool Calls

How is it calculated?

1. Collect the tool calls made by the agent.
2. Inspect the arguments/input parameters for each call.
3. Judge whether each set of arguments correctly addresses the user's task.
4. The LLM determines whether the arguments are correct.

Simple example: Task: Find AI/ML Engineer jobs in Hyderabad. The correct tool may be selected, but `role='Data Scientist'` and `location='Delhi'` are wrong arguments.

How to interpret the score: The source gives a default threshold of 0.5 and an example configuration using 0.7. Strict mode uses 1 = perfect and 0 = otherwise.

5. Task Completion (TaskCompletionMetric)

This checks whether the agent actually achieved what the user asked for. In simple words: **Did the agent finish the job?**

Formula / calculation:

Task Completion Score = AlignmentScore(Task, Outcome)

How is it calculated?

1. Extract the task from the trace, or supply it directly.
2. Extract the final outcome.
3. Compare the outcome with what the user requested.
4. An LLM-as-a-judge generates the alignment score.

Simple example: Task: Find three AI/ML jobs in Hyderabad. If the agent finds three relevant jobs and returns them, completion is high. If it searches but stops without returning them, completion is low/partial.

How to interpret the score: The source's example uses threshold 0.7. A higher score means the final outcome more completely matches the user's task.

6. Step Efficiency (StepEfficiencyMetric)

This checks whether the agent reached the result without wasting steps. In simple words: **Did it take a direct and efficient path?**

Formula / calculation:

Step Efficiency Score = AlignmentScore(Task, Execution Steps)

How is it calculated?

1. Extract the task from the trace.
2. Review all execution steps.
3. Check whether each step was necessary to complete the task.
4. Penalize unnecessary or redundant actions.
5. The efficiency score reflects how well the execution path aligns with the task.

Simple example: Efficient: Search → Filter → Rank → Return. Inefficient: Search → Search again → Search again → retrieve unnecessary information → search again → filter → rank → return.

How to interpret the score: The source's example uses threshold 0.7. Higher score means a more direct and efficient execution path.

How the Six Metrics Work Together

The source presents agent evaluation as a diagnostic process rather than a single final score. The idea is to evaluate the part of the agent responsible for a particular failure.

Agent part	Metric	Simple question
Planning	Plan Quality	Is the plan good?
Planning / execution	Plan Adherence	Did it follow the plan?
Tool selection	Tool Correctness	Did it choose the right tool?
Tool inputs	Argument Correctness	Did it give the tool the right inputs?
Execution path	Step Efficiency	Did it avoid unnecessary steps?
Final outcome	Task Completion	Did it actually complete the task?

Important distinction: Task Completion and Step Efficiency are not the same. An agent can complete a task successfully but use too many unnecessary steps (high Task Completion, low Step Efficiency). It can also be efficient but stop before completing the task (low Task Completion, potentially high Step Efficiency).

Source basis: [AI_Agent_Evaluation_Complete \(1\).pdf](#), especially the sections on Plan Quality, Plan Adherence, Tool Correctness, Argument Correctness, Task Completion, Step Efficiency, thresholds, and the final metric matrix.

DeepEval's Three Evaluation Layers

This section is added to the previous PDF without changing its existing pages. It explains the three layers shown in the source material and maps the six metrics to the layer where they evaluate the agent.

Layer	What it evaluates	Metrics
1. Reasoning Layer	The agent's planning and reasoning: whether it creates a good plan and whether it follows that plan during execution.	Plan Quality Plan Adherence
2. Action Layer	The agent's concrete tool decisions: whether it selects the right tool and supplies the right inputs/arguments.	Tool Correctness Argument Correctness
3. End-to-End Layer	The overall execution: whether the agent takes an efficient path and ultimately achieves the user's requested outcome.	Step Efficiency Task Completion

Simple way to understand the three layers

Reasoning Layer = THINK

Look at how the agent decides what it should do. Plan Quality asks whether the strategy is good. Plan Adherence asks whether the agent actually follows that strategy.

Action Layer = ACT

Look at the agent's tool actions. Tool Correctness asks whether the right tool was selected. Argument Correctness asks whether the tool received the correct inputs.

End-to-End Layer = FINISH

Look at the complete execution. Step Efficiency checks whether the agent avoided unnecessary work, while Task Completion checks whether the requested goal was actually achieved.

Which Metric Comes Under Which Layer?

Layer	Metric	Simple question
Reasoning	Plan Quality	Was the plan good?
Reasoning	Plan Adherence	Did the agent follow its plan?
Action	Tool Correctness	Did it choose the right tool?
Action	Argument Correctness	Did it give the tool the right inputs?
End-to-End	Step Efficiency	Did it avoid unnecessary actions?
End-to-End	Task Completion	Did it actually complete the task?

How the layers fit together

PLAN → TOOL → ARGUMENTS → STEPS → OUTCOME

PLAN	TOOL	ARGUMENTS	STEPS	OUTCOME
Plan Quality Plan Adherence	Tool Correctness	Argument Correctness	Step Efficiency	Task Completion

Important point from the source

The source presents these metrics as a diagnostic system. The idea is to evaluate the part of the agent responsible for a particular failure: poor planning → Plan Quality; failure to follow a good plan → Plan Adherence; wrong tool → Tool Correctness; wrong tool parameters → Argument Correctness; unnecessary actions → Step Efficiency; and failure to achieve the user's goal → Task Completion.

Source basis: AI_Agent_Evaluation_Complete (1).pdf. The mapping follows the source's Reasoning, Action, and Overall Execution/End-to-End organization.